

Week 7

Smart Home Appliance Management System

SCENARIO

A smart home company is developing software to manage different electrical appliances.

Although every appliance works differently internally, every appliance can:

- Turn On
- Turn Off
- Display Status

The company wants programmers to **hide the implementation details** while providing a **common interface** for all appliances.

INSTRUCTION

1. Login to your GitHub & select your OOP repository.
2. Create a new directory called "week_7".
3. Move into the week_7 directory & create the following java files:
 - 3.1. **Appliances.java**
 - 3.2. **Main.java**

ACTIVITY

1. Complete the code template given & the expected output.
2. Add **TWO (2)** more appliances classes as suggested below:
 - 2.1. Air Conditioner
 - 2.2. Washing Machine
 - 2.3. Microwave
 - 2.4. Television
 - 2.5. Refrigerator

```

public abstract class Appliance {

    protected String brand;

    public Appliance(String brand) {
        _____ = _____;
    }

    public void displayBrand() {
        System.out.println("Brand : " + _____);
    }

    public void turnOn() {
        System.out.println("Power ON");
    }

    public void turnOff() {
        System.out.println("Power OFF");
    }

    public abstract void _____;

}

public class main {

    public static void main(String[] args) {

        _____ =
            _____;

        _____ =
            _____;

        _____;
        _____.turnOn();
        _____;
        _____;

        System.out.println();

        _____;
        _____;
        _____.operate();
        _____.turnOff();
    }
}

```

EXPECTED OUTPUT

Brand : LG
Power ON
Washing clothes...
Power OFF

Brand : Panasonic
Power ON
Store food & beverages...
Power OFF